

Which Rural Places Move Kids Up? Neighborhood Correlates of Upward Mobility Within Rural Labor Markets

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Research on neighborhoods and intergenerational mobility is overwhelmingly urban, and the Opportunity Atlas notes that rurality helps children in some regions and hurts them in others without saying what separates the two. I ask which neighborhood characteristics distinguish rural Census tracts with above-average upward mobility from otherwise similar rural tracts. Comparing 17,310 rural tracts to other tracts inside the same commuting zone, the share of single-parent households is the largest adjusted correlate of adult income rank for children raised at the 25th percentile: a one-standard-deviation increase (8.4 percentage points) corresponds to a 1.51 percentile-point lower mean adult income rank (95% CI -1.69 to -1.34), about a quarter of a standard deviation of rural tract mobility. Local economic conditions are statistically indistinguishable from zero once a region is held fixed. Ten tract characteristics add only 0.097 to R^2 beyond commuting-zone indicators alone, which explain 0.564 on their own: most of what predicts rural mobility is which labor market a tract sits in, not the tract itself. These are descriptive conditional associations, not causal estimates.

intergenerational mobility | rural America | neighborhood effects | Opportunity Atlas

1. Introduction

Where a child grows up predicts how much they earn as an adult, even after conditioning on parental income, and part of that association is a causal exposure effect: children who move to higher-opportunity areas earlier in childhood earn more as adults, roughly in proportion to the years they spend there (1, 2). That literature is almost entirely metropolitan. Moving to Opportunity randomized vouchers in five large cities, the county-level exposure estimates are most precise where population is densest (3), and the segregation literature takes the metropolitan neighborhood as its unit of interest (8).

Rural America is largely missing from this account. The Opportunity Atlas observes that rurality is associated with higher mobility in some regions and lower mobility in others without identifying what separates the two (4). Roughly one in four tracts in the merged Atlas sample has fewer than 100 people per square mile, and place-based opportunity policy is usually designed around urban mechanisms that may not describe what varies across rural places. Growing up in a rural area, I wanted to dive deeper into how mobility works there, and challenge some of my pre-conceived notions on the topic.

This paper asks: among rural Census tracts inside the same local labor market, which neighborhood characteristics distinguish those with above-average upward mobility from those with below-average upward mobility, and do those correlates differ from the ones that operate in dense urban tracts? I answer it with tract-level Opportunity Atlas data, comparing rural tracts only against other tracts in the same commuting zone, and I report every estimate with a cluster-robust interval.

Among rural tracts in the same commuting zone, single-parent household share carries the largest adjusted association with children's adult income rank. This is larger than mean household income, poverty, employment, job growth, commute time, or third-grade math scores. Meanwhile, the local economic indicators are indistinguishable from zero. This extends the national pattern in (5), where family structure was the strongest of five candidate correlates

Significance

Research on how neighborhoods shape children's adult earnings is built almost entirely on cities, and the Opportunity Atlas notes that growing up rural helps children in some regions and hurts them in others without saying what separates the two. Comparing rural Census tracts against other tracts in the same local labor market, the share of single-parent households is the largest measured correlate of adult income rank for children raised poor, while local economic conditions show no measurable association. The effect of region, however, dwarfs that of any tract characteristic.

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M.F. designed the analysis, wrote the code, and wrote the paper. The author declares no competing interests.

across commuting zones, by showing that the same ordering survives *within* rural areas of a single labor market, where composition-based explanations for it are weakest. The rural pattern is also not a rescaled copy of the urban one: poverty share flips sign between rural and dense-urban tracts on a common scale.

2. Related work

Neighborhood effects on mobility. Chetty and Hendren estimate that neighborhoods affect children's adult outcomes through cumulative childhood exposure, using families who move between areas at different child ages (2, 3), and the Moving to Opportunity reanalysis reaches the same conclusion experimentally (1). This work identifies *that* place matters and how much; it says less about *which* features of a place track opportunity, and is estimated almost entirely on urban movers.

Correlates of mobility across places. A parallel descriptive literature asks what distinguishes high- from low-mobility places. Across commuting zones, mobility correlates with residential segregation, income inequality, school quality, social capital, and family structure, the last being the strongest single correlate (5); the Opportunity Atlas pushed this to the tract level (4). My analysis sits in this strand and extends it in two ways: by restricting attention to rural tracts, which are pooled into the national picture in that work, and by comparing tracts only within commuting zones. Rural places have been studied directly. Weber et al. (11) find upward mobility highest in noncore rural counties, and Connor et al. (12) show that the share of children raised in two-parent households accounts for nearly all of the rural–urban gap. Both, however, compare rural places with urban ones. I ask instead what separates rural tracts from *each other* inside the same labor market.

Mechanisms: schools, sorting, and race. Two of my predictors are already contested in this literature. Third-grade math scores stand in for school quality, which tracks measured achievement gaps (7, 9) but is itself something families sort on (8). And mobility differs sharply by race in ways neighborhood characteristics alone do not explain (6), so the large positive coefficient on share white is a conditional association consistent with documented national race gaps, not a place effect under this design.

3. Data

Source and time window. All data come from the Opportunity Insights Opportunity Atlas public tract files (4), downloaded in `code/00_pull.ipynb`: table 4 (`tract_outcomes_simple.csv`, 73,278 tracts, read with the seven columns used here) and table 9 (`tract_covariates.csv`, 74,044 tracts, 36 columns after dropping the commuting-zone fields duplicated across the two files). The raw files are not committed; the pull notebook reproduces them. Outcomes describe the 1978–1983 birth cohorts measured in adulthood (roughly ages 31–37, around 2014–2015) from federal income tax records. Covariates are measured in 2000 except third-grade math scores (2013) and Census mail return rate (2010).

Unit of analysis. In the original source each row is a 2010 Census tract, and each outcome is an estimate for the children who *grew up* there, aggregated across all children of parents at a given income percentile; the Atlas releases no individual records. My unit of analysis is the same. No repeated measurement arises: each tract appears once, and the outcome is a single cohort-pooled statistic, not a panel.

Outcome. `kfr_pooled_pooled_p25`: the mean percentile rank in the national adult household income distribution for children whose parents sat at the 25th percentile, pooling across child race and gender: 0.43 means those children reached the 43rd percentile on average. Because the outcome is a rank, coefficients read as percentile points once multiplied by 100.

Predictors. Ten tract characteristics, fixed before estimation to span family structure, economic conditions, human capital, social connectedness, and racial composition: share of single-parent households, share below the poverty line, mean household income, share of adults employed, annual average job growth 2004–2013, mean commute time, share with a BA or higher, mean third-grade math score, Census mail return rate, and share white. All are defined once

in `code/utils.py` (COVARIATES), so every notebook and figure uses the same names and labels.

Rural definition and analysis sample. Rural means fewer than 100 people per square mile in 2000 (`popdensity2000`): a transparent project-specific cutoff, not the Census Bureau’s official classification, which is built from block-level population and land-use criteria. Density is an imperfect proxy, though the commuting-zone fixed effects in Section 4 absorb part of this, since such a tract is compared against other tracts in the same metropolitan labor market. Table 1 characterizes the merged sample: 71,958 tracts across 741 commuting zones, 17,978 of them rural, with median tract mobility 0.425 (SD 0.071) and a median of 380 children behind each estimate.

Table 1. Analysis sample. Merged Opportunity Atlas tracts after the one-to-one join and the drop of tracts missing mobility or population density. Mobility is the mean adult household income rank of children whose parents were at the 25th percentile; the child count is the number of children behind each tract’s estimate.

Analysis sample	Value
Census tracts in analysis sample	71,958
Rural tracts (<100 people/sq. mi.)	17,978
Commuting zones represented	741
Median tract mobility (income rank)	0.425
Std. dev. of tract mobility	0.071
Median children per tract estimate	380

Missing cells. Counts of tracts with no value, per field, are in `output/table2_missing_cells.tex` in the repository. The outcome is missing for 1.6% of tracts and at the same 1.6% rate among rural tracts. Job growth has the highest overall rate (3.5%) but is *less* often missing in rural tracts (1.1%); only the Census mail return rate is appreciably worse in rural tracts (2.5% versus 0.9%). I expected suppression to fall disproportionately on sparse rural tracts, which would have made the rural sample a biased subset of rural America. It does not.

Limitations of the source. (i) The Atlas reports estimates, not populations: each tract value carries sampling error, largest where few children grew up, which is disproportionately rural, addressed below with a minimum-count screen. (ii) Covariates are cross-sectional and two are measured after the focal cohorts’ childhood years. (iii) Tracts use 2010 boundaries but the children grew up in the 1980s and 1990s. (iv) Complete-case estimation drops 668 of 17,978 rural tracts (3.7%).

4. Methods

Analysis runs in three numbered notebooks that execute in order and define their functions at the top: `00.pull.ipynb` downloads, `01.merge.ipynb` merges and cleans, `02.analyze.ipynb` estimates and writes every figure and table here. All paths resolve relative to `code/utils.py`.

Cleaning and merging. The two Atlas tables are joined on the tract key (`state`, `county`, `tract`) with an **inner join**, validated as **one-to-one** (`pandas.merge(..., how="inner", validate="one_to_one")`), so a silent many-to-many blowup would raise rather than inflate the sample. Row counts are printed before and after: 73,278 outcome rows and 74,044 covariate rows in, 73,199 matched rows out. I then drop 1,241 tracts missing either the mobility outcome or population density, leaving **71,958** tracts. This is an exact-match merge on a shared numeric key; no fuzzy matching or blocking is involved.

Derived variables. The density cutoffs are defined once in `code/utils.py`: rural below 100 and dense urban at or above 10,000 people per square mile. The child count (`pooled_pooled_count`) is carried through the merge for the screen.

Estimation. For the rural sample I estimate

$$y_{ic} = \alpha_c + \sum_{k=1}^{10} \beta_k z_{ic}^{(k)} + \varepsilon_{ic}, \quad [1]$$

where y_{ic} is mean adult income rank for tract i in commuting zone c , α_c is a commuting-zone fixed effect, and $z^{(k)}$ is characteristic k standardized to

mean zero and unit variance within the estimation sample. Standard errors are clustered by commuting zone and all intervals are cluster-robust. The fixed effects make every comparison a within-labor-market one: a coefficient answers “among rural tracts in this commuting zone, how does mobility differ between tracts one standard deviation apart on this characteristic?” rather than “how does Appalachia differ from the upper Midwest?” Standardizing puts shares, dollars, minutes, and test-score points on one axis; the cost is that a coefficient means nothing without the standard deviation it is scaled by, which Section 5 states explicitly.

Explained variance. I report the *incremental* R^2 : the R^2 of Equation 1 minus that of a commuting-zone-indicators-only model. The full-model R^2 does not measure what the tract characteristics explain, because the fixed effects already absorb most of the variance.

Precision screen. Atlas estimates resting on few children are noisy, so I refit Equation 1 on the 15,186 rural tracts (84.5%) whose estimates rest on at least 200 children. This is a sensitivity check rather than the preferred specification, since dropping the sparsest tracts changes which rural places the sample describes.

Rural versus dense-urban comparison. To ask whether the rural pattern is distinctive I fit Equation 1 separately on rural and on dense-urban tracts ($\geq 10,000$ people per square mile). Coefficients from two separately standardized models are not comparable, because one standard deviation means something different in each: the standard deviation of single-parent share is 2.2 times larger among dense-urban tracts. For this comparison only, both models are standardized on **pooled** rural-plus-dense-urban means and standard deviations, so a unit is the same quantity in both. That is why the rural single-parent coefficient is -0.021 in Figure 2 but -0.015 in Figure 1 and Table 2: same estimate, different scale. The rural-only scale is quoted for the headline result.

What this design does not identify. This is descriptive, not causal: there is no exogenous variation in tract characteristics. Commuting-zone fixed effects remove between-region composition, but not the sorting of families into tracts within a region, and not unobserved tract differences. Coefficients are conditional associations, and I use “associated with” deliberately.

5. Results

Mobility varies more within density categories than between them: median rural mobility is modestly higher than dense-urban, but the interquartile range within every category is wider than the gaps between category medians. Rurality alone is not a useful predictor of a child’s outcome, so the analysis below compares rural tracts with each other rather than with urban ones.

Family structure is the largest adjusted correlate in rural tracts. Figure 1 and Table 2 report Equation 1 estimated on 17,310 rural tracts across 727 commuting zones. Single-parent household share carries by far the largest association: a one standard deviation increase (8.4 percentage points) corresponds to a **1.51 percentile-point lower** mean adult income rank (95% CI -1.69 to -1.34 , $p < 0.001$). Rural tract mobility has a standard deviation of 6.3 percentile points, so this is roughly a quarter of a standard deviation, which is the largest single association in the model, and about a third larger in magnitude than the next-largest coefficient.

Share white is the second largest ($+1.15$ points per SD) and BA share the third ($+0.74$). The four local economic indicators are the striking null result: annual job growth ($+0.02$, $p = 0.56$), mean commute time ($+0.04$, $p = 0.48$), share of adults employed ($+0.05$, $p = 0.49$), and mean household income (-0.07 , $p = 0.60$) are all indistinguishable from zero once region is held fixed. A rural tract’s labor-demand conditions carry no measurable adjusted association with the adult incomes of the children raised there, consistent with the exposure literature’s framing of mobility as a childhood-environment process rather than a labor-market one (1).

Region explains far more than tract characteristics do. Commuting-zone indicators alone account for $R^2 = 0.564$ of the variation in rural tract mobility; adding all ten characteristics raises this to 0.661, an incremental R^2 of **0.097**. These ten characteristics therefore explain about a tenth of rural mobility variation once region is accounted for. Most of what predicts a rural

Single-parent share has the largest adjusted rural association

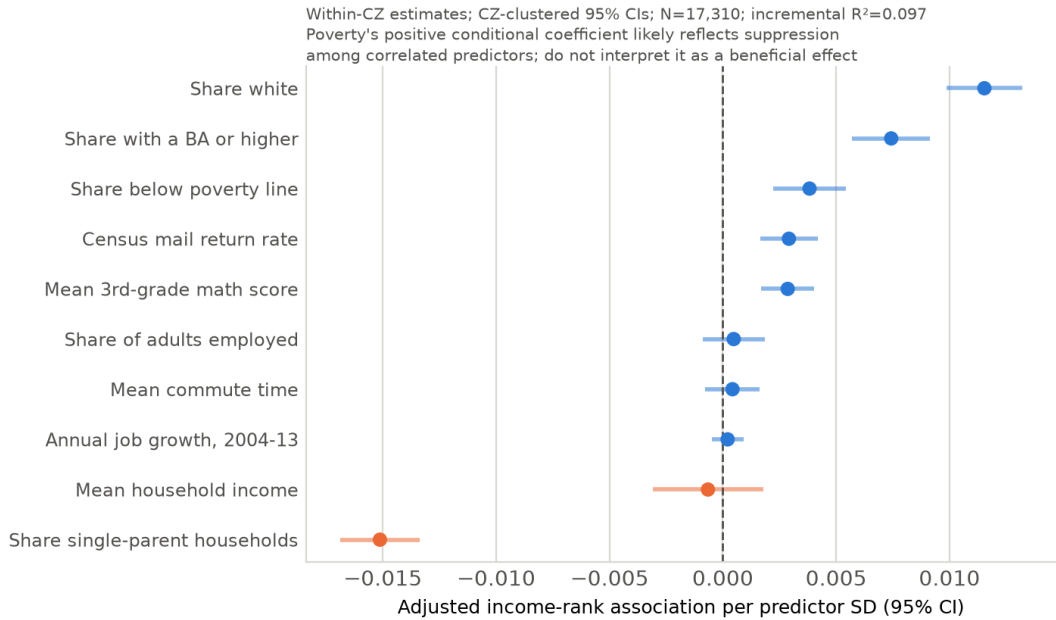


Fig. 1. Single-parent share has the largest adjusted rural association. OLS coefficients from Equation 1 on 17,310 rural tracts, expressed per standard deviation of each characteristic among rural tracts, with commuting-zone fixed effects and commuting-zone-clustered 95% confidence intervals. Blue denotes positive and orange negative associations. Job growth, commute time, employment share, and mean household income overlap zero.

Table 2. Within-commuting-zone OLS results, rural tracts. Coefficients are changes in mean adult household income rank per standard deviation of each characteristic among rural tracts; multiply by 100 for percentile points. Confidence intervals are clustered by commuting zone. $N = 17,310$ tracts in 727 commuting zones. The reported R^2 is the full-model value and is dominated by the commuting-zone indicators; see the incremental R^2 in the text.

Characteristic (per SD)	β	95% CI	p
Share single-parent households	-0.0151	[-0.0169, -0.0134]	< 0.001
Mean household income	-0.0007	[-0.0031, 0.0018]	0.599
Annual job growth, 2004-13	0.0002	[-0.0005, 0.0009]	0.558
Mean commute time	0.0004	[-0.0008, 0.0016]	0.484
Share of adults employed	0.0005	[-0.0009, 0.0019]	0.494
Mean 3rd-grade math score	0.0029	[0.0017, 0.0040]	< 0.001
Census mail return rate	0.0029	[0.0016, 0.0042]	< 0.001
Share below poverty line	0.0038	[0.0022, 0.0054]	< 0.001
Share with a BA or higher	0.0074	[0.0057, 0.0091]	< 0.001
Share white	0.0115	[0.0099, 0.0132]	< 0.001
N	17,310		
R^2	0.661		

child’s adult income rank is which regional labor market they grew up in, not which tract inside it.

The result survives a precision screen. Refitting on the 14,741 rural tracts whose estimates rest on at least 200 children (output/table4.count_sensitivity.tex) leaves the ranking intact. Single-parent share is essentially unchanged and slightly larger (−0.0151 to −0.0157), as are share white and third-grade math. Three coefficients change sign: job growth, commute time, employment share. All three are null in the main model, so these are movements of noise around zero, not unstable associations; that table marks significance so this is visible without the surrounding text.

The rural pattern is not a scaled-down urban one. Figure 2 places both groups on a common pooled standard-deviation scale. Single-parent share leads in both settings but is markedly *stronger* in dense-urban tracts (−0.031 versus −0.021 per pooled SD), and share below the poverty line *reverses sign*: positive in rural tracts (+0.005), negative in dense-urban ones (−0.012).

Mean household income does the same, near zero in rural tracts and strongly negative in dense-urban ones (−0.016). The positive rural poverty coefficient should not be read as poverty benefiting children: poverty share is heavily collinear with the other predictors ($r = 0.64$ with mean household income, 0.62 with single-parent share), so a coefficient that reverses its bivariate sign here is best understood as suppression. The comparison is **exploratory**: the dense-urban model has only 28 commuting-zone clusters, below the range where cluster-robust standard errors are reliable (10). It compares shape, not magnitude.

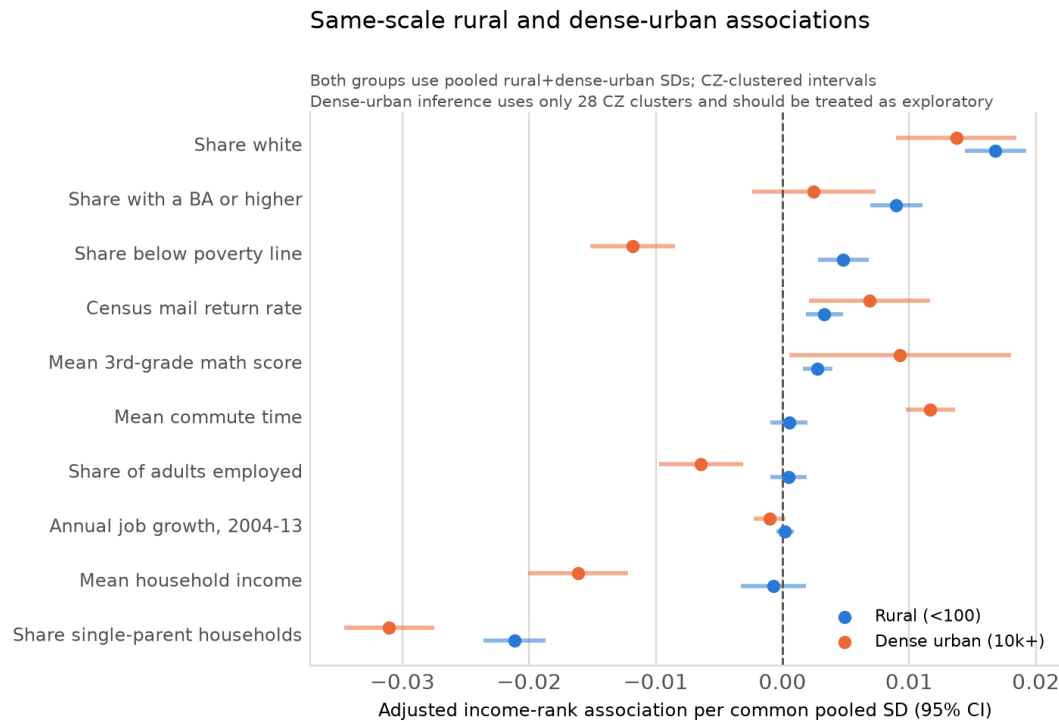


Fig. 2. Same-scale rural and dense-urban associations. Coefficients from Equation 1 fit separately on rural (<100 people per square mile, $N = 17,310$, 727 clusters) and dense-urban ($\geq 10,000$, $N = 2,468$, 28 clusters) tracts, both standardized by pooled predictor standard deviations so the two are directly comparable. Because the scale differs from Figure 1, the same rural coefficient appears here as −0.021 rather than −0.015. Dense-urban intervals rest on 28 clusters and are exploratory.

6. Discussion

Among rural Census tracts in the same commuting zone, single-parent household share is the largest adjusted correlate of the adult income rank of children raised in low-income families, and local economic conditions are indistinguishable from zero. This extends the national cross-commuting-zone pattern in (5) into rural America and into a within-labor-market comparison, and it survives a precision screen. It is also the within-rural counterpart to (12): they find that two-parent household share accounts for the rural advantage over urban places, and I find that the same characteristic orders rural tracts against one another. The second result is a caution about magnitude: ten tract characteristics add under a tenth to the variance region alone explains.

Four limitations bound these claims. They are conditional associations, not effects: the design cannot separate a neighborhood effect from families sorting into neighborhoods, and single-parent share may proxy for family resources that travel with children. The predictors are strongly collinear (mean income with BA share, $r = 0.78$), so the poverty coefficient is a suppression artifact. “Rural” is my own density cutoff, not the Census Bureau’s. And the dense-urban comparison rests on 28 clusters.

The most important finding of this project was the correlation between single-parent households and mobility, and I think future work on rural mobility could be targeted towards finding the causes of this correlation, and other traits associated with the share of single-parent households.

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787 **A. Agentic Analysis** 851

788 For this project I mainly used Claude Code, and occasionally used OpenAI 852

789 Codex to review or challenge the work that Claude had completed. My 853

790 typical workflow was to start by giving the assistant the context in detail, 854

791 including where my mind was, what I had currently completed, and the project 855

792 specifications. I also had my agents keep a markdown file with current progress 856

793 on the project and my general tastes as context for the next session. I found the 857

794 assistants particularly useful for learning new statistics concepts that I could 858

795 implement in my project. When the assistant would implement something, I 859

796 would check its work, use another assistant to challenge it, and ask it to explain 860

797 things it had implemented that were outside of my knowledge. One specific 861

798 example of something I rejected was the slide design for the presentation. My 862

799 professor noted that slide 2 was not providing important information. I agreed, 863

800 then hand-picked a new figure to put on the slide. I also took this feedback 864

801 into account when writing this paper, making sure to not miss the mark and 865

802 include information that is irrelevant. 866

803 867

804 **Transcripts.** 868

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- 806 • Claude: [agentic_transcript_fortner.docx](#) 870
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